

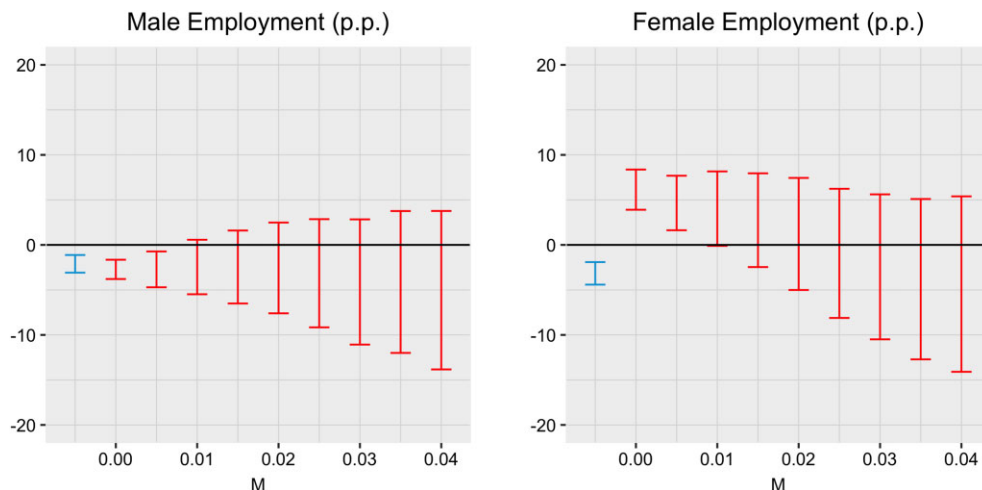


FIGURE 7
Sensitivity analysis for $\theta = \tau_{15}$ using $\Delta = \Delta^{\text{SD}}(M)$.

intervals for $\hat{\beta}_{15}$ from specification (20). In red, we plot FLCIs when $\Delta = \Delta^{\text{SD}}(M)$ for different values of M ; recall that $M = 0$ corresponds with allowing only for linear violations of parallel trends, and larger values of M allow for larger deviations from linearity. In the analysis for men (left panel), the FLCIs are similar to those from OLS when allowing for violations of parallel trends that are approximately linear ($M \approx 0$), but become wider as we allow for more non-linearity; the breakdown value for a significant effect is $M \approx 0.01$. For women (right panel), the original OLS estimates are negative and the confidence interval rules out 0. When we allow for linear violations of parallel trends ($M = 0$), however, the picture changes substantially owing to the pre-existing downward trend that is visible in Figure 6. Indeed, for $M < 0.01$ the robust confidence set contains only positive values. Intuitively, this is because the point estimate for $t = 15$ lies above a linear extrapolation of the negative pre-trend. Thus, if we were to impose the same smoothness restrictions for men as for women, we would either have to reconcile significant effects of opposite signs by gender (if $M < 0.01$) or we would not be able to rule out null effects for both genders ($M \geq 0.01$).

How can we interpret the magnitudes of M in this example? We consider a calibration exercise based on the magnitudes of possible possible confounds: if violations of parallel trends were driven by confounding changes in education quality, what would a given value of M imply about the evolution of those confounds? Chetty *et al.* (2014) estimate that a 1 standard deviation increase in teacher value-added (VA) corresponds with a 0.4 percentage point increase in adult employment. Hence, a value of $M = 0.01$ would correspond with allowing the slope of the differential trend to change by the equivalent of a one-fourtieth of a standard deviation of teacher VA across consecutive periods. Since the robust confidence sets for both men and women begin to include zero around this value of M , the strength with which we can rule out a null effect depends on our assessment of the economic plausibility of such non-linearities.

Finally, in [Supplementary Material, Appendix C](#), we again compare the reported confidence sets to the estimated identified set $\mathcal{S}(\hat{\beta}, \Delta)$. Our results suggest that sampling uncertainty is large relative to the length of the identified set when M is close to zero, but becomes less important as M increases.

7. CONCLUSION

This paper considers the problem of conducting inference in DiD and related designs that is robust to violations of the parallel trends assumption. We introduce a variety of restrictions on the class of possible differences in trends that formalize commonly made arguments in empirical work, generalizing the framework for partial identification in [Manski and Pepper \(2018\)](#). We provide inference procedures that are uniformly valid so long as the difference in trends satisfies these restrictions, and derive novel results on the power of these procedures. We recommend that applied researchers report robust confidence sets under economically motivated restrictions on parallel trends. We also recommend that researchers conduct formal sensitivity analyses, in which they report confidence sets for the causal effect of interest under a variety of possible restrictions on the underlying trends. Such sensitivity analyses make transparent what assumptions are needed in order to draw particular conclusions.

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Supplementary Data

[Supplementary data](#) are available at *Review of Economic Studies* online.

Data Availability Statement

The data and code underlying this research is available on Zenodo at <https://doi.org/10.5281/zenodo.7388015>.

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